

Development of a Modular In-ear Wearable Device for Dementia Monitoring: A Preliminary Approach

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Abstract. HowMI is a modular in-ear wearable device in early development, designed to enable continuous health monitoring for individuals living with dementia. The device aims to non-invasively measure key vital signs, including heart rate and body temperature, using integrated photoplethysmography and infrared sensors. Design requirements were defined through a systematic literature review and organized into physical, emotional, and cognitive domains, guided by user-centered and inclusive design principles. Based on these requirements, a compact hardware architecture was established, featuring an ESP32 microcontroller and Bluetooth communication for real-time data transmission. A physical prototype is currently under development to test sensor integration and evaluate potential configurations for optimal placement within the ear canal. The HowMI device is conceived as part of a broader modular system, which includes a wrist-worn unit and an AI-based platform for data processing and caregiver support. These initial developments lay the groundwork for future ergonomic studies, sensor performance analysis, and user validation.

Keywords: Wearable Device · Dementia · Vital Signs · Inclusive Design

1 Introduction

Globally, more than 55 million people live with dementia [7], and an even greater number of caregivers are also affected by the disease. As the condition progresses,

the caregiver’s role becomes increasingly complex, with physical and emotional exhaustion, isolation, and depression being common[5]. Wearable technology can offer valuable support in monitoring this health condition. This research continues the development of a wearable device called “HowMI - Home Wearables and Monitors Integrated”, designed to enhance end-user confidence by providing healthcare professionals with an innovative and more accurate care management tool, enabling faster responses in emergency situations [2]. This short paper presents the technological approach, the preliminary studies, and the future directions of the device in the context of dementia care.

2 Methodology

As part of a broader system under development, two types of wearable devices are being created: an In-Ear Wearable Device (HowMI) [2] and a Wrist Wearable Device [8, 9], each is designed to monitor specific physiological and behavioral parameters. While the overall system integrates both devices, this paper focuses specifically on the preliminary development of the HowMI.

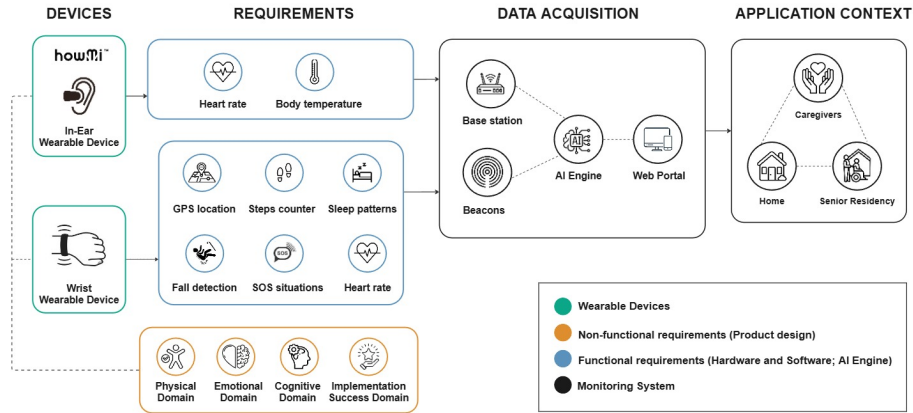


Fig. 1. Overview of the project system.

To support this system-level approach, the overall architecture is illustrated in Figure 1. It highlights the integration between the wearable devices and the supporting technological infrastructure. Data collected by the devices is transmitted wirelessly to base stations or beacons, then relayed to a central artificial intelligence (AI) engine [1]. This engine performs real-time data analysis, enabling anomaly detection and generating alerts. Processed information is securely made available through a web portal accessible to caregivers and healthcare professionals. The modular nature of the system ensures that each device

can operate independently or in combination, allowing flexible deployment in different care settings, such as private homes or senior residential facilities.

The development of HowMI followed an iterative process guided by user-centered and inclusive design principles, with an emphasis on safety, usability, and efficiency for elderly individuals with dementia. Given the early stage of development, direct involvement of end users was deferred. Instead, initial design requirements were derived from a systematic literature review [3], which synthesized the common needs, preferences, and challenges faced by individuals with dementia and their caregivers. Hardware and software development proceeded in parallel. The current prototype of HowMI integrates temperature and heart rate sensors, selected for their relevance in early detection of physiological changes and their suitability for continuous, non-invasive monitoring in dementia care contexts.

3 Preliminary Studies

In this section, we present the preliminary studies that served as the foundation for our project development.

3.1 Product Design

To support this research, a systematic literature review was undertaken to explore current trends and challenges in the use of wearable technology for dementia care. The review underscored the emerging potential of ear-based devices such as smart earphones and hearing aids in monitoring key physiological indicators, including heart rate, oxygen saturation, and body temperature. In response to identified limitations in existing solutions, a conceptual design framework was developed, based on user-centered and universal design principles. The resulting requirements were organized into three core domains: physical, emotional, and cognitive. Furthermore, the specific requirements for the HowMi device emphasize the importance of a user-friendly interface, broad adaptability across users, and continuous real-time health data monitoring to enable timely intervention [3].

3.2 Hardware

As a first objective in validating the HowMi device, the development and optimisation of the previously designed in-ear device were carried out, with particular emphasis on the hardware. The aim was to reduce the physical dimensions of the device and address any potential limitations without compromising the accuracy of essential biometric data measurements, such as body temperature and oxygen saturation.

As a result of this optimisation, it was possible to reduce the overall length by approximately 1.84 mm, the maximum width by 1.25 mm, and the thickness by 0.6 mm. Two sensors were used, arranged perpendicularly: one infrared sensor,

MLX90632[6], positioned to face the eardrum in order to measure body temperature, and a second photoplethysmography (PPG) sensor, MAX30102[4], placed in contact with the ear canal to measure heart rate and oxygen saturation.

At this stage, a state-of-the-art review was also conducted to identify the most recent sensory technologies applied in wearable devices, with a particular focus on in-ear solutions. To ensure both relevance and up-to-date information, only articles published from 2019 onwards were selected, written in English and with full free access. The inclusion criteria targeted studies addressing technologies related to the measurement of body temperature, blood oxygen saturation, or heart rate.

This review enabled the exploration of the currently available sensory technologies and a better understanding of their limitations. Two technologies were identified for measuring body temperature in the ear canal: infrared and contact-based. Infrared sensors are limited in that incorrect alignment with the ear canal may lead to inaccurate readings, while contact-based technology requires larger devices that fully occupy the ear canal.

For heart rate and oxygen saturation, photoplethysmography (PPG) technology was observed, which requires contact with the skin for measurement. A notable limitation of this method is that minor movements may interfere with the readings, leading to measurement errors.

The sensors communicate with an ESP32 microcontroller via the I²C protocol, and the device transmits the data via Bluetooth to a mobile phone.

4 Future Work

Future work will focus on advancing the development of the HowMI in-ear wearable device through a series of targeted technical and usability enhancements. A key next step involves conducting a detailed anatomical study of the human ear using 3D models reconstructed from computed tomography (CT) scan data. This analysis will provide critical measurements to optimize the ergonomic fit of the device and ensure user comfort, particularly for long-term use by individuals with dementia. In parallel, a prototype is being developed to support initial testing of the integrated sensors. This prototype will allow the evaluation of sensor performance in an in-ear environment, with particular attention to accuracy, signal stability, and sensitivity to movement. Comparative testing of different sensing technologies and configurations will be conducted to identify the most suitable solutions in terms of precision, energy efficiency, and mechanical compatibility with the ear canal. Further efforts will also address the integration of the in-ear device within the broader HowMI system architecture. This includes refining data transmission protocols, enhancing interoperability with the wrist-worn device, and ensuring seamless communication with the central AI-based data processing platform.

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